HACKATHON PROPOSAL

Samvād

Your AI News Companion That Speaks Your Language

Voice-First Contextual News Intelligence

Hackathon	Mistral Worldwide Hackathon 2026
Track	API Track (AWS Bedrock)
Location	Online
Team	Devashish + Shreyash
Date	March 1–2, 2026

Built as part of the Minerva / Samāchār ecosystem

1. Executive Summary

The Problem

News disengagement isn't a content problem — it's a context problem. Readers lack the background knowledge to understand articles that assume prior knowledge. But most of the day, people can't even read: they're commuting, cooking, exercising. There's no way to consume and deeply understand news without a screen.

The Solution: Samvād

A voice-first AI news companion that briefs you on today's top stories and lets you naturally ask "wait, who is that?" or "why does that matter?" in conversation. It combines Mistral Large 3 for intelligence, Pixtral Real-Time for speech recognition, and ElevenLabs for natural voice output. No screen required.

The Name

Samvād (संवाद) means "dialogue" or "conversation" in Hindi/Sanskrit. It captures the core interaction: not just consuming news, but conversing with it, understanding it, and building knowledge through discussion.

Samvād is a vertical slice of the broader **Minerva / Samāchār** vision — extending contextual news intelligence from screen to voice. If it works, this becomes a second modality for the platform.

2. Product Vision & User Journey

2.1 Target User

Primary persona: A 22–35 year old commuter who wants to stay informed but finds traditional news intimidating or inaccessible during their daily routine. They're walking, on transit, cooking, or exercising — their hands and eyes are busy, but their ears are free.

2.2 Core Interaction Modes

Mode	User Says	AI Does
Morning Briefing	"What's happening today?" or "Brief me on tech news"	Reads 3–5 top stories as conversational summaries, pausing between each
Context on Demand	"Who is that?" or "What's the ECB?"	Explains the entity/concept with news-relevant context via RAG (Wikipedia/Wikidata → Mistral)
Deep Dive	"Tell me more" or "Why does that matter?"	Expands with broader context, related events, and implications

Follow-up Q&A	"How does this affect the UK?" or "What happened before this?"	Answers using article context + retrieved knowledge, maintains conversation history
Article Mode	Pastes/shares an article URL	Parses article, identifies key entities, offers to walk through or answer questions about it

2.3 Conversation Flow Example

User	"Hey, what's happening in Europe today?"
Samvād	"Good morning! Here are today's top stories. First, the European Central Bank has signalled a potential rate pause at its March meeting, breaking from its recent cutting cycle. ECB President Christine Lagarde cited persistent services inflation..."
User	"Wait — what's the ECB exactly?"
Samvād	"The European Central Bank is the central bank for the eurozone — the 20 EU countries that use the euro. It sets interest rates and monetary policy for the region, similar to what the Federal Reserve does in the US. Lagarde has been its president since 2019. Want me to continue with the story, or dive deeper into the ECB?"
User	"Continue with the story."
Samvād	"So the rate pause matters because the ECB had cut rates three times since last June. Markets were expecting a fourth cut, so this signals the ECB thinks inflation isn't fully under control yet. Moving on — second story: the UK government announced a new AI safety framework..."
User	"How does the ECB rate decision affect the UK?"
Samvād	"Good question. The UK isn't in the eurozone — it has its own central bank, the Bank of England. But ECB decisions still matter because a stronger euro from higher rates can make UK exports more competitive. Also, the Bank of England often considers ECB policy when making its own decisions. As I mentioned, the ECB had been cutting while the Bank of England held steady..."

Notice how the AI: references what it explained earlier ("As I mentioned"), connects stories together, and adapts depth based on questions.

3. Technical Architecture

3.1 High-Level System Design

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                              SAMVĀD ARCHITECTURE

  User (Voice)                        Mobile Web UI
      │                                     │
      │  speak                       visual feedback
      ▼                                     ▼
┌──────────────────────────────────────────────────┐
│           FRONTEND (Next.js)                     │
│  ┌──────────────┐  ┌──────────┐  ┌──────────┐    │
│  │ Pixtral      │  │Waveform  │  │11 Labs   │    │
│  │ STT Client   │  │+ Trans-  │  │TTS Stream│    │
│  │ (WebSocket)  │  │ cript    │  │(WebSocket)│    │
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└──────────────────────────────────────────────┘
      │                                  ▲
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┌──────────────────────────────────────────────┐
│          BACKEND (FastAPI)                   │
│                                              │
│  ┌──────────────────────────────────────┐    │
│  │ Conversation Manager                 │    │
│  │ - Session memory (what's been discussed) │
│  │ - Intent detection (briefing/context/QA) │
│  │ - Entity tracking per session        │    │
│  └──────────────────────────────────────┘    │
│        │            │            │            │
│        ▼            ▼            ▼            │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐    │
│  │News      │  │Entity    │  │Mistral   │    │
│  │Ingestion │  │Context   │  │Large 3   │    │
│  │(RSS)     │  │(RAG)     │  │(Bedrock) │    │
│  └──────────┘  └──────────┘  └──────────┘    │
│      │           │                           │
│      ▼           ▼                           │
│  ┌──────────┐  ┌──────────────┐              │
│  │RSS Feeds │  │Wikipedia +   │              │
│  │(Europe)  │  │Wikidata APIs │              │
│  └──────────┘  └──────────────┘              │
└──────────────────────────────────────────────┘
```

3.2 Tech Stack

Layer	Technology	Rationale
STT	Pixtral Real-Time (Mistral)	Free, stays in Mistral ecosystem, real-time WebSocket
LLM	Mistral Large 3 (via AWS Bedrock)	Best Mistral model for reasoning, $200 credits via API track
TTS	ElevenLabs Streaming API	Best voice quality, streaming reduces perceived latency, targets special prize
Backend	FastAPI (Python 3.11)	Async-first, fast, great for ML pipelines
Frontend	Next.js 14 + TypeScript	SSR, great mobile support, Vercel deployment
Entity Context	Wikipedia + Wikidata APIs	Free, reliable, comprehensive knowledge base
News Sources	European RSS feeds	BBC, Reuters, Guardian, FT, DW — relevant to judges
Session State	In-memory (dict per session)	Simple for MVP, no DB needed for hackathon
Deployment	Vercel (frontend) + Railway (backend)	Free tiers, fast deployment

3.3 News Sources (Europe-Focused)

Source	Category	RSS Feed URL

BBC News	General	feeds.bbci.co.uk/news/rss.xml
BBC Europe	Europe	feeds.bbci.co.uk/news/world/europe/rss.xml
Reuters World	General	feeds.reuters.com/reuters/worldNews
The Guardian	General	theguardian.com/world/rss
Deutsche Welle	Europe	rss.dw.com/rdf/rss-en-all
France 24	Europe	france24.com/en/rss
TechCrunch	Tech	techcrunch.com/feed/
Ars Technica	Tech	feeds.arstechnica.com/arstechnica/index
The Verge	Tech	theverge.com/rss/index.xml
Nature	Science	nature.com/nature.rss

3.4 Entity Context Pipeline (RAG)

When a user asks about an entity, the system follows this pipeline:

Step	Action	Details
1	**Intent Detection**	Mistral classifies: is this a briefing request, entity question, follow-up, or article mode?
2	**Entity Extraction**	If entity question: extract entity name from user utterance using Mistral function calling
3	**Session Check**	Check if entity was already explained in this session → reference prior explanation
4	**Wikidata Lookup**	Search Wikidata API for entity ID, basic facts, dates, relationships
5	**Wikipedia Extract**	Fetch 2-paragraph extract from Wikipedia for background
6	**Context Assembly**	Combine: article context + Wikidata facts + Wikipedia extract + conversation history
7	**LLM Generation**	Mistral Large 3 generates 2–3 sentence explanation, optimized for spoken delivery
8	**TTS Streaming**	Response streams to ElevenLabs for immediate voice output

4. Core Prompt Templates

4.1 System Prompt

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You are Samvād, a voice-first AI news companion. You help people understand the
news through natural conversation.

PERSONALITY:
• Warm, knowledgeable, like a well-read friend explaining the news over coffee
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• Concise by default — expand only when asked
• Honest about uncertainty: "I'm not sure about that" is always acceptable
• Neutral on controversial topics — present multiple perspectives

VOICE OPTIMIZATION:
• Use short sentences (max 20 words). Spoken language is different from written.
• Avoid parenthetical asides, em dashes, or complex clause structures.
• Use conversational transitions: "So," "Now," "Here's the thing," "Moving on,"
• Pause naturally between stories: end with a question or transition.
• Numbers: say "about 3 billion" not "$2,987,654,321"
• Acronyms: spell out on first use, then use the short form

SESSION MEMORY:
• Reference entities you've already explained: "As I mentioned earlier..."
• Connect stories: "This relates to what we discussed about..."
• Track what the user knows: don't re-explain unless asked

CONTEXT DELIVERY:
• When explaining an entity: 2-3 sentences focused on why it matters NOW
• When briefing: summarize each story in 3-4 sentences, pause for questions
• When answering follow-ups: be direct, then offer to go deeper
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4.2 Intent Classification Prompt

Used as a function call to route user input to the right handler:

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Classify the user's intent into exactly one category:

BRIEFING — User wants a news summary ("what's happening", "brief me", "top
stories")
ENTITY_QUERY — User asks about a person/org/concept ("who is", "what's the",
"explain")
FOLLOW_UP — User asks a deeper question about current topic ("why", "how", "tell me
more")
ARTICLE_MODE — User shares a URL or asks about a specific article
NAVIGATION — User wants to move on ("next story", "skip", "go back")
CHITCHAT — Greeting or off-topic ("hello", "thanks", "goodbye")

Return JSON: {"intent": "...", "entity": "..." (if ENTITY_QUERY), "topic": "..."
(if BRIEFING)}
```

4.3 Entity Context Prompt

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Explain "{entity_name}" for a listener who just heard it in a news story.

ARTICLE CONTEXT: {surrounding_text}
WIKIDATA FACTS: {wikidata_facts}
WIKIPEDIA EXTRACT: {wikipedia_extract}
PREVIOUSLY DISCUSSED: {session_entities}

RULES:
1. 2-3 sentences optimized for SPOKEN delivery (short, clear, no jargon)
2. Focus on why this entity matters to the current news story
3. If we already discussed a related entity, reference it naturally
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4. End with a brief offer: "Want to know more, or should I continue?"
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5. Latency Management Strategy

Latency is the biggest risk for a voice-first experience. Here's the mitigation plan:

Technique	How It Works	Impact
Streaming TTS	ElevenLabs starts speaking as tokens arrive from Mistral — don't wait for full response	Perceived latency drops from ~2s to ~500ms
Pre-fetch Briefings	Fetch and summarize top stories on app load, before user asks	First briefing is instant
Short System Prompt	Keep system prompt under 500 tokens; move context to user messages	Faster time-to-first-token
Concise Responses	System prompt enforces 2–3 sentence answers unless asked for more	Less generation time, more natural for voice
Entity Cache	Cache Wikipedia/Wikidata lookups in memory for session duration	Second query about same entity is instant
Filler Phrases	While RAG fetches, play brief filler: "Let me think about that..."	Covers 1–2s gap naturally
Parallel Fetching	Fetch Wikidata + Wikipedia simultaneously using asyncio.gather()	Cuts retrieval time in half

6. Frontend Design

The UI is intentionally minimal — the voice IS the interface. The screen provides visual feedback and fallback.

6.1 Mobile-Friendly Layout

Top Bar	Samvād logo + current topic/story indicator + settings icon
Main Area	Live waveform animation (listening/speaking states) + current transcript
Entity Cards	Scrollable row of entity chips for entities mentioned in conversation — tap to ask about any
Input Area	Large microphone button (push-to-talk or voice-activated) + text fallback input
Status Bar	Connection status + "Samvād is thinking..." / "Listening..." / "Speaking..."

6.2 Visual States

State	Visual	Audio
Idle	Subtle breathing animation on mic button	Silent
Listening	Waveform active (blue), mic pulsing, live transcript appearing	STT active
Thinking	Waveform paused, "thinking" animation, entity chips loading	Optional filler phrase
Speaking	Waveform active (green), transcript of AI response appearing	TTS playing
Error	Red pulse on mic, brief error message	"Sorry, I didn't catch that"

7. Task Breakdown & Division of Labor

Team

Devashish: ML/AI pipeline, Mistral integration, RAG, prompts, news ingestion, conversation logicShreyash: Backend API structure, frontend voice interface, WebSocket handling, TTS/STT integration, deployment

7.1 Hour-by-Hour Sprint Plan

SATURDAY (Hours 0–12)

Time	Milestone	Devashish	Shreyash
10–11 AM	**Setup & Scaffold**	Create GitHub repo, FastAPI skeleton, env setup, claim Mistral + AWS credits	Next.js scaffold, Tailwind + shadcn setup, basic mobile layout
11 AM–1 PM	**Core Pipeline V1**	RSS fetcher for 5 European sources, basic article parsing + summarization via Mistral	Pixtral STT WebSocket client, browser audio capture, basic mic button UI
1–2 PM	**Lunch + Sync**	Integration check: can backend receive text and return a response?	Integration check: can frontend capture audio and display transcript?
2–4 PM	**RAG Pipeline**	Wikipedia + Wikidata API integration, entity context retrieval, intent classification prompt	ElevenLabs TTS streaming integration, audio playback, visual waveform
4–6 PM	**Conversation Loop**	Conversation manager: session memory, entity tracking, multi-turn context	Connect STT → Backend → TTS full loop, WebSocket for streaming

6–7 PM	**Dinner + AWS Session**	Attend AWS Strands Agents live coding (7 PM Paris) — learn multi-agent patterns	Continue frontend polish
7–10 PM	**Integration Sprint**	End-to-end voice conversation working: briefing + entity questions + follow-ups	Entity chips UI, visual states (listening/thinking/speaking), mobile responsiveness
10 PM–12 AM	**Edge Cases**	Test with 10+ real articles, tune prompts for spoken delivery, handle errors gracefully	Error states, loading animations, connection handling, push-to-talk refinement

SATURDAY NIGHT: Go to sleep. Seriously.

SUNDAY (Hours 13–24)

Time	Milestone	Devashish	Shreyash
8–9 AM	**Fresh Start**	Re-test full pipeline with fresh news, fix overnight breakages	Deploy backend to Railway, frontend to Vercel, test production
9–11 AM	**Article Mode**	Add article URL parsing: user pastes URL → AI walks through it conversationally	Article URL input in UI, entity chips auto-populate from article entities
11 AM–1 PM	**Polish Sprint**	Prompt tuning: make responses more natural, concise, conversational. Test latency.	UI polish: animations, transitions, mobile testing on real phone
1–2 PM	**Lunch + Demo Prep**	Write demo script: planned conversation showing all modes	Record backup demo video in case live demo fails
2–3 PM	**Final Testing**	Run through demo script 3x, fix any issues	Final deploy, README, GitHub cleanup, screenshots
3–4 PM	**Submission**	Submit on hackathon platform: description, GitHub link, demo video	Final check: everything works on production URLs

8. Scope Management

This is the most critical section. The difference between winning and crumbling is knowing what to cut.

8.1 Must Have (MVP)

- Voice conversation loop: STT → Backend → TTS working end-to-end

- Morning briefing mode: summarize 3–5 top European stories
- Entity context on demand: "Who is X?" → RAG → spoken explanation
- Session memory: AI references what it already explained
- Mobile-friendly web UI with mic button and transcript
- Deployed and accessible via public URL

8.2 Should Have (If Time Permits)

- Article URL mode: paste a link, AI walks through it
- Entity chips: visual list of discussed entities, tap to re-ask
- Waveform visualization for listening/speaking states
- Filler phrases during RAG retrieval
- Follow-up Q&A connecting multiple stories

8.3 Won't Do (Explicitly Out of Scope)

- User authentication or accounts
- Persistent storage or database (beyond in-memory session)
- Multilingual support (English only for MVP)
- Fine-tuning any models
- Native mobile app
- Personalized knowledge profiles
- Publisher integrations

9. Risk Mitigation

Risk	Likelihood	Impact	Mitigation
High voice latency (>3s)	Medium	High — kills UX	Streaming TTS, pre-fetched briefings, filler phrases, concise prompts
Pixtral STT quality issues	Medium	Medium	Fallback: browser Web Speech API (free, works offline, zero setup)
AWS Bedrock rate limits	Low	High	Use Mistral API directly as fallback ($15 free credits)
ElevenLabs credits run out	Low	Medium	Fallback: browser speechSynthesis API (lower quality but free)
RSS feeds fail/change	Low	Low	Hardcode 3 backup article JSONs for demo reliability

Scope creep (Sunday AM)	High	High	Strict feature freeze at 11 AM Sunday — only bug fixes after
Live demo fails at pitch	Medium	High	Record backup demo video by 2 PM Sunday

10. Pitch & Demo Strategy

10.1 The 5-Minute Pitch Structure

Time	Section	Content
0:00–0:30	**Hook**	"Raise your hand if you've ever read a news article and had to Google a name or term to understand it. Now imagine you're commuting — you can't Google anything. That's the problem we're solving."
0:30–1:30	**Live Demo**	Live voice interaction: ask for briefing → get 2 stories → ask "who is that?" → get instant context → ask follow-up. The demo IS the pitch.
1:30–2:30	**Problem**	News disengagement is a context problem. 68% of young adults feel excluded by news that assumes knowledge they don't have. Most of their day, they can't even use a screen.
2:30–3:30	**Solution**	Samvād: a voice-first news companion. Brief me, explain that, tell me more. Three simple interactions that make news accessible anywhere.
3:30–4:30	**Architecture**	Quick tech overview: Pixtral STT → Mistral Large 3 (Bedrock) → RAG (Wikipedia/Wikidata) → ElevenLabs TTS. All Mistral-native. Session memory for coherent conversations.
4:30–5:00	**Vision**	This is a slice of Minerva — our broader platform for contextual news intelligence. Voice is the second modality. This hackathon validated the interaction model.

10.2 Demo Script

Pre-plan a conversation that reliably showcases all features:

1. "Hey Samvād, what's happening in Europe today?" → Shows briefing mode
2. (After hearing about ECB) "What's the ECB?" → Shows entity context via RAG
3. "How does that affect the UK?" → Shows follow-up reasoning + session memory
4. "Next story" → Shows navigation
5. (Share article URL) "Walk me through this article" → Shows article mode (if implemented)

11. Special Prize Strategy

Prize	How We Qualify	Strength

ElevenLabs Best Voice AI	Core product is voice-first; ElevenLabs TTS is central to the experience, not a bolt-on	Very strong — voice is the primary interface, not an add-on feature
Mistral Best Use of Mistral	Mistral Large 3 for reasoning + Pixtral Real-Time for STT = dual Mistral model usage	Strong — using two Mistral products together in a novel way
Startup Prize (if applicable online)	Samvād extends Minerva, a real startup with 40+ waitlist, 15+ user interviews, accelerator applications	Strong — this isn't a toy project, it's validating a real product direction

12. Success Criteria

12.1 Definition of Done (Sunday 4 PM)

- A user can open the web app, tap the mic, and have a voice conversation about today's news
- The AI can brief on 3–5 stories, explain entities, and answer follow-up questions
- Session memory works: the AI references prior explanations naturally
- Latency is under 3 seconds from user finishing speaking to AI starting to respond
- Deployed on public URLs (Vercel + Railway)
- GitHub repo is public with clear README
- Demo video recorded as backup
- Submitted on hackathon platform with description and links

12.2 Stretch Goals

- Article URL mode working
- Entity chips in UI
- Waveform animations polished
- Multi-story connection ("This relates to what we discussed about...")
- Public votes on hackathon platform

> *"The best hackathon projects are ones where you see them on social media and go 'wow, this is really cool' without needing 10 minutes to understand it."*
>
> — Joffrey Thomas, Mistral DevRel

Let's build something people want to talk to. 🎙️